Jaisidh Singh

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Education

2024–present (expected 2026) **MSc in Machine Learning**, *Eberhard-Karls Universität Tübingen*, Tübingen.
- German GPA: 1.52/5.0
- Thesis (with OpenEuroLLM: Aaron Klein & Antonio Orvieto): scaling behaviour of hybrid LLMs
- Previous semesters: published 1× at EMNLP Main Conference & 2× at ICLR Workshops.

2020–2024 **B.Tech in AI & Data Science**, *Indian Institute of Technology*, Jodhpur.
- Thesis: negation-aware compositional reasoning in vision-language models.

Awards & fellowships: Konrad Zuse School (ELIZA) Fellowship for Master's Students in Germany 2024-2026.

Work/Research Experience

Apr 2026 – present **Thesis Researcher at OpenEuroLLM**, *With Dr. Aaron Klein & Dr. Antonio Orvieto*.
- **Improved data efficiency by** $\approx 30\%$ in LLM pretraining with hybrid-attention architectures.
- **Developed an efficient distributed training backend** for scaling LLM pretraining, including throughput profiling, cost analysis, & experiment management **to optimize GPU-hours by** 20% over naive grid-search.

Oct 2025 – May 2026 **SWE Graduate Assistant at Tübingen AI Center**, *With Prof. Peter Gehler*.
- Designed & **deployed to production real-time interactive computer vision systems** for artistic stylization of user images using semantic segmentation and pose estimation based warping.
- Developed & delivered a hands-on workshop on **agent orchestration with LangGraph at GCPR 2025**, covering tool-use, memory, & MCP towards a **deep-research agent**.

Jan 2025 – May 2025 **ML Research Assistant at ELLIS Institute Tübingen**, *With Dr. Antonio Orvieto*.
- **Reduced VLM training cost by** 90% for modality stitching, by using a hypernetwork-based framework. Results verified for CLIP & LLaVA-style VLMs across ViT, T5, GPT-2,Qwen, & Pythia backbones. **Published: EMNLP 2025 Main Conference**.

Winter 2022 & 2023 **Undergraduate Researcher at Trusted AI Lab IITJ**, *With Prof. Mayank Vatsa, Prof. Richa Singh*.
- Designed a compositional negation benchmark & fine-tuned CLIP on custom objectives, **achieving SOTA accuracy on negation understanding** and $\approx 4\%$ **mean performance improvement** across diverse retrieval tasks. **Published: WACV 2025 (50 citations)**.
- Systematically analysed StyleGAN2 latent spaces to profile identity leakage via dual-space identity invariance, informing privacy-preserving generative AI guidelines. **Published: WACV 2024**.

Summer 2022 & 2023 **Research Engineer Intern at Bosch Research India**, *With Dr. Amit A. Kale & Sonam Singh*.
- **Developed an interpretable failure-discovery system** for semantic segmentation models for industry-scale autonomous driving data (BDD, A2D2, ACDC), **achieving** 95% **precision** in automatically detecting human-coherent errors via clustering. **Technical report: arXiv:2411.10845**.
- **Created multi-modal image retrieval pipelines** using transformers for interpretable evaluation of weather effects on camera feeds for autonomous vehicles.

Selected Publications

2025 **(Almost) Free Modality Stitching of Foundation Models**, ***EMNLP 2025, ICLR-W 2025***,
Jaisidh Singh, Diganta Misra, Boris Knyazev, Antonio Orvieto.

2025 **Learning the Power of "No": Foundation Models with Negations**, ***WACV 2025***,
*Jaisidh Singh**, Ishaan Shrivastava*, Mayank Vatsa, Richa Singh, Aparna Bharati.

(Refer to my Google Scholar profile for a complete list of publications. * indicates equal contribution.)

Technical Skills

- **Machine learning:** PyTorch, JAX, FLAX, Haiku, Optax, Huggingface, Triton
- **Distributed training & HPC:** Slurm, Megatron-LM, Data/Tensor Parallelism
- **Agentic tools:** dsPy, n8n, LangGraph, LangChain, Claude Code, OpenCode
- **SWE tools:** Git, GitHub, CI/CD, Kubernetes, Flyte, Prometheus
- **Programming languages:** Python, C++, Bash, JavaScript, Dart